Generalized Three-Layer Integrated Interleaved Codes

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***Abstract*— Integrated interleaved (II) codes allocate parities sharable by codewords, which are also called interleaves, and hence enable a good tradeoff on locality and correction capability. Better locality translates to shorter decoding latency, which is essential to many new applications, such as failure recovery in distributed storage. The shared parities can be allocated in multiple layers to improve the locality. However, existing work only generalized two-layer II codes to allow multiple interleaves with extra errors to be corrected. For three or more layers, the shared parities in a layer only provide extra protection over one of the interleaves or subgroups they cover in the next lower layer. This letter proposes a construction of generalized three-layer II codes. In our codes, the shared parities can be used to correct multiple interleaves in each subgroup and the allocation of their correction power among the interleaves is flexible. Our scheme corrects much more error patterns than the prior three-layer II codes, and achieves better locality than the two-layer GII codes.**

***Index Terms*— Erasure codes, generalized integrated interleaved codes, locally recoverable, three-layer.**

I. Introduction

ERROR-CORRECTING codes are adopted in numerous digital storage and communication systems. To meet the high-speed and low-latency requirements of next-generation applications, shorter codes are desired. Without sacrificing the target reliability, the overall redundancy can be reduced by adopting parities shared by codewords to correct extra errors. These codes can be naturally used for local failure recovery, which is essential to enabling the continued scaling of high-performance distributed storage. The better the locality, which is defined as the number of symbols to access for decoding, the shorter the latency.

In integrated interleaved (II) codes [1], a set of linear block codewords, also called interleaves, are nested to create codewords of a stronger linear block code that is a subcode of the base linear block code. Unlike other codes with shared parities, the II codes do not need another code to protect the shared parities. The II codes are generalized in [2] so that the codewords generated from the nesting can have different correction capabilities. The partial maximum distance separable (PMDS) codes [3] allow maximum flexibility on the usage of the shared parities to correct extra errors. Nevertheless, their explicit construction is only available for special cases, and large finite fields are needed. Shared parities are also defined using parity check matrices in the generalized concatenated (GC) codes [4], which have higher implementation complexity than the GII codes [2] when the error correction capability is not small. All these codes are two-layer codes. The first layer is the base code, and the second layer consists of the codes generated by nesting *all* the interleaves. If any interleave has extra errors, *all* interleaves are accessed to utilize the shared parities. To achieve better locality, the shared parities can be allocated to subgroups of interleaves in a hierarchical manner in multiple layers. When there are fewer extra errors, the interleaves in a subgroup and the corresponding shared parities are sufficient for decoding. A special class of the II scheme with multiple layers was introduced in [5]. In this scheme, a shared parity can only provide extra protection for one of the interleaves or subgroups it covers in the next lower layer. Less significant locality improvement is achievable for erasure-only case by the GC codes with a modified decoding process [4] and the modified GII codes with a lower-zero-triangular nesting matrix [6], both of which are two-layer codes.

This paper proposes a generalized construction of three-layer II codes. Compared to the special case in [5] for three layers, our scheme is generalized in two dimensions: i) Multiple levels of codes are allowed in layer two and three, and hence multiple interleaves (subgroups) with extra errors are correctable. It should be noted that the 'level' in this paper refers to the number of codes used in a layer; ii) The shared parities from layer three allow the correction of multiple additional interleaves. Our construction is done through analyzing the required properties of the nesting matrices in order to correct multiple interleaves from multiple subgroups. Examples show that our three-layer GII codes correct much more error patterns than the three-layer II codes in [5], and hence achieve better performance.

This paper is organized as follows. Section II introduces the two-layer GII codes. The proposed three-layer GII codes are detailed in Section III. Comparison results are provided in Section VI, and conclusions follow in Section V.

II. Two-Layer GII Codes

The GII scheme [2] works on a set of m interleaves of length n denoted by $c_0, c_1, \cdots, c_{m-1}$. Each interleave belongs to the first-layer code $\mathcal{C}_0(n, k_0)$. Usually Reed-Solomon codes over $GF(q)$ are considered. The second layer involves υ codes $\mathcal{C}_i(n, k_i)$ $0 < i \leq \upsilon < m$ such that $\mathcal{C}_\upsilon \subseteq \mathcal{C}_{\upsilon-1} \subseteq \cdots \subseteq \mathcal{C}_0$. A two-layer GII code is defined as

$$\mathcal{C} = \Big\{ c = [c_0, c_1, \cdots, c_{m-1}] : c_i \in \mathcal{C}_0, \\ \tilde{c}_j = \sum_{i=0}^{m-1} \alpha^{ij} c_i \in \mathcal{C}_{\upsilon-j}, 0 \leq j < \upsilon \Big\}$$

where α is a primitive element of $GF(q)$. The nesting in the above definition is also expressed as

$$\begin{bmatrix} \tilde{c}_0 \\ \tilde{c}_1 \\ \vdots \\ \tilde{c}_{\upsilon-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \alpha & \alpha^2 & \cdots & \alpha^{m-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{(\upsilon-1)} & \alpha^{2(\upsilon-1)} & \cdots & \alpha^{(\upsilon-1)(m-1)} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{m-1} \end{bmatrix}. \quad (1)$$

Manuscript received November 1, 2017; accepted November 30, 2017. Date of publication December 6, 2017; date of current version March 8, 2018. The associate editor coordinating the review of this paper and approving it for publication was C. Feng. *(Corresponding author: Xinmiao Zhang.)*

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Digital Object Identifier 10.1109/LCOMM.2017.2780132

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In this paper, a vector and its polynomial representation are used interchangeably. Let the received (corrupted) interleaves be $y_i(x) = c_i(x) + e_i(x)$ $(0 \le i < m)$, where $e_i(x)$ is the error vector. $|e_i|$ denotes the hamming weight of $e_i(x)$. The error-correction capability of $\mathcal{C}_i$ is $t_i = (n - k_i)/2$, and $t_0 < t_1 \le \cdots \le t_\upsilon$. Sort $|e_i|$ to τ_i such that $\tau_0 \le \tau_1 \le \cdots \le \tau_{m-1}$. The decoding will be successful iff $\tau_{m-1-\upsilon+i} \le t_i$ for $i = 0, 1, 2, \cdots, \upsilon$.

$2t$ valid syndromes are needed to correct t errors. In the decoding, if $|e_i| \le t_0$, then the errors are correctable within the interleave itself making use of $2t_0$ syndromes $S_j^{(i)} = e_i(\alpha^j) = y_i(\alpha^j)$ $(1 \le j \le 2t_0)$. Higher-order syndromes are not computable from $y_i(x)$, since $c_i \in \mathcal{C}_0$ and hence $c_i(\alpha^j) \ne 0$ and $e_i(\alpha^j) \ne y_i(\alpha^j)$ for $j > 2t_0$. Instead, if any interleave has $|e_i| > t_0$ errors, then the extra syndromes needed are obtained from the nested syndromes defined as $\tilde{S}_j^{(i)} = \tilde{y}_i(\alpha^j)$, where $\tilde{y}_i(x) = \sum_{j=0}^{m-1} \alpha^{ij} y_j(x)$. Since $\tilde{c}_i(x) = \sum_{j=0}^{m-1} \alpha^{ij} c_j(x) \in \mathcal{C}_1$ for $0 \le i < \upsilon$, $\tilde{c}_i(\alpha^j) = 0$ for $2t_0 + 1 \le j \le 2t_1$. As a result, let the indices of the $b \le \upsilon$ failed interleaves be $l_1, l_2, \cdots, l_b$, their $2(t_1 - t_0)$ higher-order syndromes can be derived through

$$\left[S_j^{(l_1)}, S_j^{(l_2)}, \cdots, S_j^{(l_b)}\right]^T = A^{-1}\left[\tilde{S}_j^{(0)}, \tilde{S}_j^{(1)}, \cdots, \tilde{S}_j^{(b-1)}\right]^T, \tag{2}$$

where A is the $b \times b$ submatrix formed by the first b rows and columns $l_1, l_2, \cdots, l_b$ in the nesting matrix of (1). After the interleaves with at most t_1 errors are corrected, the nested syndromes are updated. Then the $2t_2 - 2t_1$ higher syndromes for the remaining uncorrected interleaves are derived using an equation similar to (2). This process is repeated to derive higher-order syndromes for the interleaves with more errors.

III. Three-Layer GII Codes

In this paper, interleaves that have more than t_0 errors are called exceptional interleaves. In the two-layer GII scheme explained in Section II, all interleaves need to be accessed if any of them is exceptional. Reading and transferring more symbols cause longer latency and larger network traffic overhead in a distributed system. One effective way to improve the locality is to group the interleaves and apply the nesting in multiple layers. When there are fewer extra errors to be corrected, the nestings in lower layers involving fewer interleaves are utilized. Let $m = m_1 m_2$. The only three-layer II scheme is defined in [5] for a special class as follows.

$$\mathcal{C} = \begin{cases} c = [c_0, c_1, \cdots, c_{m-1}] : c_i \in \mathcal{C}_0, \\ \sum_{i=0}^{m_2-1} c_{jm_2+i} \in \mathcal{C}_1,\ 0 \le j < m_1, \quad \sum_{i=0}^{m-1} c_i \in \mathcal{C}_2 \end{cases}$$

In the above construction, $\upsilon = 1$ in layer two and three. Moreover, the parities generated through $\mathcal{C}_2$ in layer three only provide extra protection for one interleave if there is no other exceptional interleave in the same layer-two subgroup, which include c_{jm_2+i} with the same j. More explanation on this aspect is provided in the following.

Next, we first analyze the conditions need to be satisfied in order to utilize the parities in the nested layers to correct extra errors. Then a construction is proposed for a generalized three-layer GII scheme, in which $\upsilon > 1$ is allowed in both layers and the layer-three parities can be used to correct additional interleaves besides those allowed by the layer-two nesting.

As shown in (1), a nesting scheme is defined by a matrix. To correct extra errors in the interleaves, additional syndromes are acquired through transforming the nested syndromes, such as by using (2) for the two-layer case. Since the transformation matrix A in (2) is a Vandermonde matrix regardless of which interleaves are exceptional and hence which columns of the nesting matrix in (1) are taken, it is always non-singular. Therefore, extra interleave syndromes can be always derived from the nested syndromes if $\tau_{m-1-\upsilon+i} \le t_i$ $(0 \le i \le \upsilon)$.

Now consider three-layer nesting in which the m interleaves are divided into m_1 subgroups of m_2 interleaves. In our scheme, the same nesting, defined by matrix G_1, is applied to each subgroup in layer two. The third-layer nesting, whose matrix is G_2, is applied to all interleaves. Assume that G_1 and G_2 have υ_1 and υ_2 levels of codes, respectively. The dimensions of G_1 and G_2 are $\upsilon_1 \times m_2$ and $\upsilon_2 \times m$, respectively. The syndromes from the G_1 nesting can be used to correct exceptional interleaves in the same subgroup according to a similar method as for the two-layer GII scheme. If the errors exceed the correction capability of a subgroup, the syndromes from the G_2 nesting are utilized. Denote the nested syndromes from the second layer by $\tilde{S}_j^{(i)}(0 \le i < m_1\upsilon_1)$, and those from the third layer are $\hat{S}_j^{(i)}(0 \le i < \upsilon_2)$. Use superscript (p, q) to represent the column vector formed by the syndromes whose (i) are in the range of $[p, p + q - 1]$. For example, $S_j^{(3,5)} = [S_j^{(3)}, \cdots, S_j^{(7)}]^T$. The interleave syndromes are connected to the nested syndromes through

$$\left[\begin{array}{c|c|c|c} G_1 & 0 & \cdots & 0 \\ \hline 0 & G_1 & \cdots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \cdots & G_1 \\ \hline \multicolumn{4}{c}{G_2} \end{array}\right] \left[\begin{array}{c} S_j^{(0,m_2)} \\ \hline S_j^{(m_2,m_2)} \\ \hline \vdots \\ \hline S_j^{(m-m_2,m_2)} \end{array}\right] = \left[\begin{array}{c} \tilde{S}_j^{(0,\upsilon_1)} \\ \hline \tilde{S}_j^{(\upsilon_1,\upsilon_1)} \\ \hline \vdots \\ \hline \tilde{S}_j^{((m_1-1)\upsilon_1,\upsilon_1)} \\ \hline \hat{S}_j^{(0,\upsilon_2)} \end{array}\right]. \tag{3}$$

The first matrix in (3) is referred to as the joint nesting matrix Γ. When there are exceptional interleaves beyond the correction capability of the G_1 nesting, pick the columns in Γ corresponding to all exceptional interleaves. Assume there are b such columns. The errors are only correctable if there is a non-singular $b \times b$ square submatrix in those columns. It accordingly serves a similar purpose as the A matrix in (2) to transform the nested syndromes to interleave syndromes.

For the special case of the three-layer II codes in [5], both G_1 and G_2 are row vectors of '1's. Hence, the columns of the joint nesting matrix that correspond to the interleaves in the same subgroup in layer two are identical. If more than one column is taken from a subgroup, the resulted columns would not have a non-singular square submatrix that can be used to transform the syndromes. Therefore, at most one exceptional interleave from each subgroup is correctable. This is already enabled by the G_1 nesting. The contribution of the G_2 nesting

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is to allow more errors in one of those interleaves to be corrected if a stronger code is used in layer three.

Next a generalized three-layer II scheme is proposed to enable more flexible and efficient usage of the syndromes from the third-layer nesting. Allowing all third-layer nested syndromes to be used on any subgroups faces the same problem as in the construction of the PMDS codes. Instead, it is assumed that the third-layer nesting is designed to correct extra interleaves from $r \leq m_1$ subgroups. The target is to allow s_{r-1} extra interleaves to be corrected in one subgroup, s_{r-2} extra interleaves to be corrected in one of the remaining $m_1 - 1$ subgroups, $\cdots$, s_0 extra interleaves to be corrected in one of the remaining $m_1 - (r-1)$ subgroups. Note that $\sum_{i=0}^{r-1} s_i = \upsilon_2$. Without loss of generality, assume that $s_{r-1} \geq s_{r-2} \geq \cdots \geq s_0$. The joint nesting matrix achieving this goal is shown in (4).

$$\Gamma = \left[\begin{array}{c|c|c|c} G_1 & 0 & \cdots & 0 \\ 0 & G_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & G_1 \\ \hline G_2^{r-1} & G_2^{r-1} & \cdots & G_2^{r-1} \\ G_2^{r-2} & \alpha^{-1} G_2^{r-2} & \cdots & \alpha^{-(m_1-1)} G_2^{r-2} \\ \vdots & \vdots & \ddots & \vdots \\ G_2^{0} & \alpha^{-(r-1)} G_2^{0} & \cdots & \alpha^{-(m_1-1)(r-1)} G_2^{0} \end{array}\right]. \quad (4)$$

In (4), G_1 has the same format as the nesting matrix in (1), except that m and υ are replaced by m_2 and υ_1, respectively. The lower parts form G_2 that defines the third layer nesting. For $0 \leq i \leq r-1$,

$$G_2^i = \begin{bmatrix} 1 & \alpha^{\upsilon_1} & \alpha^{2\upsilon_1} & \cdots & \alpha^{\upsilon_1(m_2-1)} \\ 1 & \alpha^{\upsilon_1+1} & \alpha^{2(\upsilon_1+1)} & \cdots & \alpha^{(\upsilon_1+1)(m_2-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{\upsilon_1+s_i-1} & \alpha^{2(\upsilon_1+s_i-1)} & \cdots & \alpha^{(\upsilon_1+s_i-1)(m_2-1)} \end{bmatrix}. \quad (5)$$

A similar matrix as in (4) is used as the parity check matrix for the two-layer GC codes in [4]. Instead of defining the second-layer codes resulted from the concatenation, it specifies the third-layer nesting in our scheme. The third-layer nesting provides extra syndromes to $\hat{r} \leq r$ subgroups. Sort the numbers of exceptional interleaves in those $\hat{r}$ subgroups to n_i such that $n_0 \leq n_1 \leq \cdots \leq n_{\hat{r}-1}$. The decoding may be successful only if $n_i - \upsilon_1 \leq s_{i+r-\hat{r}}$ for $i = 0, 1, \cdots, \hat{r}-1$ and the errors from each of the other $m_1 - \hat{r}$ subgroups are correctable within themselves. The $\hat{r}$ corresponding block columns in the G_2 parts of (4) can be re-ordered according to $n_{\hat{r}-1}, \cdots, n_1, n_0$. After the re-ordering, the α coefficients multiplied to G_2^i still form a Vandermonde matrix. The lower triangle in the Vandermonde matrix can be reduced to zero through Gaussian elimination. From (5), G_2^i consists of the first s_i rows of G_2^{r-1}. Hence, the same row operations adopted in the Gaussian elimination can be applied to reduce the lower triangular blocks in the re-ordered G_2 parts of (4) to zero. After row permutations, the reordered Γ matrix is transformed to an upper triangular format, and the blocks of matrices in the diagonal are G_1 padded below by G_2^i. From (5), these matrices are always Vandermonde matrices. Therefore, taking from them any $b \leq \upsilon_1 + s_i$ columns corresponding to the exceptional interleaves and any b consecutive rows in those columns form a (scaled version of) square Vandermonde matrix. This proves that any possible syndrome transformation matrix is invertible and extra syndromes can be derived for up to $\upsilon_1 + s_0, \upsilon_1 + s_1, \cdots, \upsilon_1 + s_{\hat{r}-1}$ interleaves in those $\hat{r}$ subgroups. It should be noted that no column/row permutation or Gaussian elimination is needed to form the syndrome transformation matrix. A similar method is used to prove the erasure correction capability of the GC codes in [4].

In our proposed scheme, although the numbers of additional interleaves can be corrected in the subgroups, namely $s_0, s_1, \cdots, s_{r-1}$, are determined in the construction of the G_2 parts, the number of extra errors correctable in an interleave is not decided by how many exceptional interleaves there are in the same subgroup. This is because that the interleave syndromes are connected to the nested syndromes by Γ. Through the multiplication of the syndrome transformation matrix, which is a square submatrix of Γ, a nested syndrome contributes to the syndromes of all interleaves that participate in the nesting. Denote the codes used in the third-layer nesting by $\mathcal{C}_i'$ and the corresponding correction capability by t_i' ($i = 1, 2, \cdots, \upsilon_2$). To be able to correct additional exceptional interleaves, the only constraint on t_i' is that each of them should be larger than t_0. Comparatively, in the three-layer II scheme of [5], t_i' must be larger than t_{υ_1}, the largest of the second-layer codes. Otherwise, the third-layer nesting does not provide any extra protection. Without loss of generality, assume that $t_0 < t_1' \leq t_2' \leq \cdots \leq t_{\upsilon_2}'$. Following a similar iterative decoding process as described in Section II, errors are correctable using the proposed three-layer nesting if the following conditions are satisfied.

- $\hat{r} \leq r$ subgroups have errors that are not correctable within each of themselves. Sort the numbers of exceptional interleaves in these subgroups to n_i such that $n_0 \leq n_1 \leq \cdots \leq n_{\hat{r}-1}$. $n_i - \upsilon_1 \leq s_{i+r-\hat{r}}$ for $0 \leq i < \hat{r}$.
- From each of those $\hat{r}$ subgroups, pick the υ_1 exceptional interleaves whose numbers of errors are closest to but not larger than $t_1, t_2, \cdots, t_{\upsilon_1-1}$. Collect the rest exceptional interleaves in these subgroups to form a set $\mathcal{I}$. Hence, $|\mathcal{I}| = \hat{\upsilon}_2 \leq \sum_{i=0}^{r-1} s_i = \upsilon_2$. Sort the numbers of errors in the interleaves in $\mathcal{I}$ to $\tau_1' \leq \tau_2' \leq \cdots \leq \tau_{\hat{\upsilon}_2}'$. $\tau_i' \leq t_{i+\upsilon_2-\hat{\upsilon}_2}'$ for $1 \leq i \leq \hat{\upsilon}_2$.

Compared to the three-layer II code in [5], our proposed scheme is generalized in two dimensions: i) More than one level of codes are allowed in layer two and three. ii) The shared parities from layer three allow the correction of multiple additional interleaves from multiple subgroups. As a result, our proposed scheme can correct much more error patterns and hence achieve better error-correcting performance. The invertibility of the syndrome transformation matrix in the proposed scheme is guaranteed based on that G_0 padded below with G_2^i is always a Vandermonde matrix, and hence its square submatrices formed by consecutive rows are always non-singular. Similar constructions of the third-layer nesting are not applicable to the modified two-layer GII codes [6], because its G_0 is zero in the lower triangle and

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TABLE I
CORRECTABLE EXTRA ERROR PATTERN COMPARISONS IN EXAMPLE ONE

Pattern	GII	II [5]	Pattern	GII	II [5]
$\{1,0 \mid 0,0\}$	yes	yes	$\{1,0 \mid 1,0\}$	yes	yes
$\{1,1 \mid 0,0\}$	yes	no	$\{1,1 \mid 1,0\}$	yes	no
$\{2,0 \mid 0,0\}$	yes	yes	$\{2,0 \mid 1,0\}$	yes	yes
$\{2,1 \mid 0,0\}$	yes	no	$\{2,1 \mid 1,0\}$	yes	no

the corresponding syndrome transformation matrices are not always non-singular. Similarly, the proposed scheme can not be extended to four or more layers.

Intuitively, three-layer GII encoding can be done by extending the two-layer GII encoding in [1] and [7]. Linear feedback shift registers generate the codewords in $\mathcal{C}_i$, and pre- and post-processing implemented as constant matrix multiplications accommodate the nesting. The overhead compared to a three-layer II encoder lies in the pre- and post-processing. Since υ_1 and υ_2 are usually small, such overhead is small when the interleaves are not very short. Compared to the three-layer II decoding, the difference in the three-layer GII decoding is the multiplication of the inverse of a non-trivial syndrome transformation matrix. Its complexity relative to that of the Berlekamp-Massey algorithm used for finding the error locator and magnitude polynomials from the syndromes is low, especially for small υ_1 and υ_2 and reasonably-long interleaves.

IV. CORRECTABLE ERROR PATTERN COMPARISONS

Using example cases, this section compares the correctable error patterns of the proposed three-layer GII scheme and those of the three-layer II codes in [5].

First consider the case that both of the three-layer GII and II schemes have $m = 4$, $m_1 = 2$, $\upsilon_1 = \upsilon_2 = 1$, $t_1 = t_0 + 1$, and $t_1' = t_0 + 2$. In another word, the four interleaves are divided into two subgroups, and layer two and three each has one single level of code. The extra correction capabilities of the layer two and three codes beyond that of the base code is one and two, respectively. Although the same codes are involved and the same amount of redundancy is employed, the proposed GII code adopts a different nesting for the third layer. The patterns of extra correctable errors are listed in Table I, in which the order of the subgroups and the order of the interleaves within each subgroup are not differentiated. The parities from the third-layer nesting in the II code can correct additional errors in a single interleave if there is no other exceptional interleaves in the same subgroup. Comparatively, the third-layer parities in our GII scheme can be used on any interleave in a subgroup. As a result, our scheme corrects errors with much more patterns and achieves better performance. Moreover, if t_1' is changed to $t_0 + 1$, the third-layer parities in our scheme can still correct one more exceptional interleave with one extra error. However, the II scheme does not benefit from the third-layer nesting in this case since $t_1' = t_1$.

In the second example, a three-layer II scheme is compared to a GII scheme with multiple levels of codes but the same amount of redundancy. For both codes, $m = 9$ and $m_1 = 3$. In the II code, $\upsilon_1 = \upsilon_2 = 1$ $t_1 = t_0 + 2$, $t_1' = t_0 + 3$. For the GII code, $\upsilon_1 = \upsilon_2 = 2$, $t_1 = t_2 = t_0 + 1$, $s_0 = s_1 = 1$, $t_1' = t_0+1$ and $t_2' = t_0+2$. Therefore, the second and third layer nestings generate 12 and 6 extra parities, respectively, in each scheme. All error patterns correctable by one scheme but not the other are listed in Table II. It can be seen that the proposed GII scheme is better at correcting more exceptional interleaves with fewer extra errors in each interleave. Hence, it achieves better error-correcting performance unless the bursts of errors are longer and there are fewer of them.

TABLE II
CORRECTABLE EXTRA ERROR PATTERN COMPARISONS IN EXAMPLE TWO

Extra error patterns correctable by the three-layer II code but not the GII code	$\{2,0,0 \mid 2,0,0 \mid 0,0,0\}\{2,0,0 \mid 2,0,0 \mid 1,0,0\}$
	$\{2,0,0 \mid 2,0,0 \mid 2,0,0\}$
	$\{3,0,0 \mid 0,0,0 \mid 0,0,0\}\{3,0,0 \mid 1,0,0 \mid 0,0,0\}$
	$\{3,0,0 \mid 1,0,0 \mid 1,0,0\}\{3,0,0 \mid 2,0,0 \mid 0,0,0\}$
	$\{3,0,0 \mid 2,0,0 \mid 1,0,0\}\{3,0,0 \mid 2,0,0 \mid 2,0,0\}$
Extra error patterns correctable by the three-layer GII code but not the II code	$\{1,1,0 \mid 0,0,0 \mid 0,0,0\}\{1,1,0 \mid 1,0,0 \mid 0,0,0\}$
	$\{1,1,0 \mid 1,0,0 \mid 1,0,0\}\{1,1,0 \mid 1,1,0 \mid 0,0,0\}$
	$\{1,1,0 \mid 1,1,0 \mid 1,0,0\}\{1,1,0 \mid 1,1,0 \mid 1,1,0\}$
	$\{1,1,1 \mid 0,0,0 \mid 0,0,0\}\{1,1,0 \mid 1,0,0 \mid 0,0,0\}$
	$\{1,1,1 \mid 1,0,0 \mid 1,0,0\}\{1,1,0 \mid 1,1,0 \mid 0,0,0\}$
	$\{1,1,1 \mid 1,1,0 \mid 1,0,0\}\{1,1,0 \mid 1,1,0 \mid 1,1,0\}$
	$\{2,1,0 \mid 0,0,0 \mid 0,0,0\}\{2,1,0 \mid 1,0,0 \mid 0,0,0\}$
	$\{2,1,0 \mid 1,0,0 \mid 1,0,0\}\{2,1,0 \mid 1,1,0 \mid 0,0,0\}$
	$\{2,1,0 \mid 1,1,0 \mid 1,0,0\}\{2,1,0 \mid 1,1,0 \mid 1,1,0\}$
	$\{2,1,1 \mid 0,0,0 \mid 0,0,0\}\{2,1,1 \mid 1,0,0 \mid 0,0,0\}$
	$\{2,1,1 \mid 1,0,0 \mid 1,0,0\}\{2,1,1 \mid 1,1,0 \mid 0,0,0\}$
	$\{2,1,1 \mid 1,1,0 \mid 1,0,0\}\{2,1,1 \mid 1,1,0 \mid 1,1,0\}$
	$\{2,1,1 \mid 1,1,1 \mid 0,0,0\}\{2,1,1 \mid 1,1,1 \mid 1,0,0\}$
	$\{2,1,1 \mid 1,1,1 \mid 1,1,0\}$

V. CONCLUSIONS

A three-layer GII scheme is proposed in this paper. Compared to the previous II codes, the proposed scheme enables more flexible usage of the redundancies achieved through the nestings. Multiple levels of codes are allowed in layer two and three. More importantly, the redundancy from the third layer can be used to correct additional exceptional interleaves. Unlike the II codes, the GII scheme does not require the codes in the third layer to be more powerful than those in the second layer. This gives more flexibility on the redundancy allocation among the codes and layers. Future research will be directed to developing more efficient decoding methods for the three-layer GII codes.

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